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Mapping Current Use of Artificial Intelligence in Pharmacology Education via a Scoping Review

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ABSTRACT

Pharmacology education, often reputed as complex, overtly didactic and decontextualised, may benefit from artificial intelligence-supported strategies. However, current guidance is fragmented across disciplines and contexts, thus weakening evidence-based curricular implementation. This scoping review mapped existing research to identify applications, strengths, limitations, and areas for future development. A double-blinded screening process facilitated by Covidence yielded 17 eligible studies from four databases. Studies mostly comprised cross-sectional studies from high-income countries in the medical context, with generative artificial intelligence being predominant (ChatGPT-3.5 and ChatGPT-4.0). Research comprised assessment ($n = 12$), pedagogy ($n = 4$), curriculum design ($n = 1$), and programme evaluation ($n = 1$). Most studies assessed tools' ability to answer examinations, with mixed success depending on the version, question type, and inclusion of context in prompt engineering. Few studies incorporated students, limiting insights into learning impact. Zero-shot prompting was mostly used, limited further by unclear design frameworks, which may bias outcomes considering downstream inefficiencies. Current research prioritises the performance of artificial intelligence, rather than its integration or impact in learning, which reduces its applicability for curriculum design and competency development. Although promising, the impact is limited, requiring clearer instructional design and rationalisation within the education ecosystem. Although there is considerable potential for pharmacology education, research requires greater structure, longitudinal design, and incorporation of students to inform clear impact. Purposeful, context-aligned implementation and continuous evaluation are needed to ensure ethical, valid, and meaningful use in pharmacology education. To support future research, recommendations are provided for practical reporting, scientific design, and impact measurement.

1 | Introduction

Pharmacology is a core component of various health professions, including dentistry [1–3], medicine [4–14], nursing [15, 16], and pharmacy [17–19]. However, the discipline has been described as suffering from a complex, content-dense nature promoting high intrinsic cognitive load [5, 15, 16], precipitating poor clinical competency development [20]. Although core pharmacology concepts have been identified to understand the discipline [16, 21], its relevance to health professions

is guided by the graduate outcome [22], necessitating clear education design to promote its direct clinical applicability [15]. Strategies, however, are often didactic and decontextualised, leading to constrained educational ecosystems that struggle to promote achievement of competence, further precipitating poor academic alignment and streamlining of approaches. Technology-enhanced learning is routinely applied in education to promote curriculum and instructional design and development [23–25], thus innovating learning modalities in parallel to changing generations, and supporting educators

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with their day-to-day processes. Artificial intelligence is one such innovation which has gained considerable attention as of late [26], spreading into various sectors, including higher education. Artificial intelligence, a broad field encompassing, among others, generative artificial intelligence, machine learning, and natural language processing, has showed diverse applicability, new reference [27] which allows for an exciting platform for educators to promote learning, overcome educational concerns in their disciplines, and promote workplace efficiency.

Considering the challenges in pharmacology education, artificial intelligence has been espoused as a potential contributor to successful learning [28]. However, as the field is still relatively new in the context of everyday-use, much is still unknown about its impact on learning, particularly in specific disciplines. Artificial intelligence, most often in the context of generative artificial intelligence, has shown use in a variety of educational functions, including the generation of learning material [29], supporting student learning [5, 23], facilitating assessment [30–32], and providing personalized feedback or remedial support [9, 29, 32]. Several guidelines are available to help support artificial intelligence's use in education [26, 31, 33, 34], however, focus is generally in the context of overarching health profession, and not necessarily the disciplines itself, such as pharmacology. Although many educational approaches are extrapolatable, applicability within specific disciplines may not be easily achieved without a unique understanding of disciplinary nuances. Given many educators' limited experience with artificial intelligence or best educational practice, the risk of poor implementation within pharmacology education increases, which potentiates the possibility of irrational educational design, reduced constructive alignment, or minimising the learning quality of the educational ecosystem. Educators may have great expectations of artificial intelligence [26], though lack sufficient evidence to support their educational design, leading to drawing from diverse contexts for local application that may not be supported through scholarship. Considering this, the study mapped the current landscape of artificial intelligence use in pharmacology education to highlight educational strengths and limitations, and recommend ways to bolster its practical use and future educational research.

2 | Methods

The Joanna Briggs Institute (JBI) 2020 guidelines for scoping reviews were used, with reporting via the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) checklist [35]. The project is registered under Open Science Framework (<https://osf.io/9t6f2>). The guiding question for the scoping review was: *How has artificial intelligence been applied in pharmacology education?*

Free text keywords and Medical Subject Heading (MeSH) terms (where available) were combined using Boolean operators (AND and OR) to compile a search string (File S1), focusing on synonyms and associated terms for “pharmacology”, “artificial intelligence” and “education”. An initial search of five relevant seeding articles was used to refine keyword selection. EBSCOHost Medline Index, PubMed, Scopus and Web of Science

were used to source literature. Primary, original research was considered eligible when it discussed artificial intelligence use in pharmacology education (regardless of the profession or discipline it was taught in) and was full-text, English articles. No date range limit was applied up to the date of conducting the scoping review (13 February 2025). Citations were excluded if they were not in English, took the form of abstract-only publications, review articles, case reports, clinical studies, or matters unrelated to pharmacology or artificial intelligence use in education. Similarly, publications that assessed the perspective or attitude towards artificial intelligence in education were excluded. Perceptual data, although important within their scope and relevant to technology uptake, were excluded to reduce conceptual heterogeneity, which could shift the focus from implementation practice to acceptability, unless such perception was grounded in implemented use.

All citations were uploaded into Covidence, with automatic duplication removal applied. Double-blinded review of titles and abstracts was performed in alignment to the eligibility criteria, after which disagreements were discussed to reach consensus. Consensus discussions were mediated by discussing each citation of concern in relation to the eligibility criteria, with assessment of alignment until both reviewers were in agreement of the final decision. Full-text articles were sourced for all eligible citations, after which a similar review process was followed, with associated exclusion reasons logged throughout the process. Inter-rater agreement was calculated as the Cohen-kappa measurement. A data extraction and quality assessment form (JBI Analytical Cross-sectional Study Checklist) was designed in Covidence to facilitate capturing relevant information, including standard characterization information, artificial intelligence model used, prompting strategies, students of interest, domain of use, and primary findings. Data was analysed and reported narratively and in tables according to domains of use.

3 | Results and Discussion

3.1 | Publication Characteristics

A total of 17 full-text sources (Table 1) were selected from the 391 citations for data extraction (Figure 1). Exclusion was predominantly due to either assessing the wrong outcome (e.g., unrelated to artificial intelligence or pharmacology) or being the wrong study design (e.g., literature review or opinion piece). Fair agreement was noted between reviewers (Cohen- κ 0.37337) due to the interpretive subjectivity of the citations' positioning of (i) artificial intelligence in education, (ii) the focus on pharmacology itself, or (iii) opinions rather than primary research. In certain cases, papers lacked initial clarity of whether artificial intelligence was applied educationally (i.e., as a tool, pedagogy or approach) or rather merely for perceived use in education (i.e., attitudes towards use or potential purposes). Similarly, broad use was often noted, negating the stratification for pharmacology itself, which would confound the interpretative focus.

Publication characteristics are presented in Table 2. Most studies were conducted in high-income countries: United States of America ($n=5$), Taiwan ($n=3$), and individual representation from Bahrain, Germany, New Zealand, Norway, Singapore,

TABLE 1 | Eligible full-text publications.

Authors, year and country	Title	Aim of study ^a	Study design and discipline or profession	Inclusion of students in experimentation, and total number of participants or samples	Focus of study	Artificial intelligence tool	Findings of the study
Bicknell et al. 2024 United States of America	ChatGPT-4 omni performance in USMLE disciplines and clinical skills: Comparative analysis	To assess and compare how GPT-3.5, GPT-4, and GPT-4 Omni perform across United States Medical Licensing Examination disciplines, clinical clerkships, and diagnostic/management skills	Cross-sectional study Medicine	No 750 clinical vignette-based multiple-choice questions across USMLE disciplines, clinical clerkships, and management clinical skills	Assessment	Zero-shot ChatGPT 3.5 ChatGPT 4.0 ChatGPT 4o	For disciplines, generative AI outperformed medical students. Where students' average was 59.3%, higher scores were present for ChatGPT 3.5 (60.6%), GPT-4.0 (81.1%) and GPT-4o (90.4%). Pharmacology scores were 57.9% versus 61.3%, 84.1% and 93.2%, respectively. GPT-4.0 and GPT-4o performed significantly ($p < 0.05$) better than GPT 3.5. Although not specific to pharmacology, clinical assessment questions followed a similar trend with 61% for medical students versus 58.9%, 81.9% and 90.9%, respectively. Generative AI was more likely to have incorrect answers for high difficulty questions ($p < 0.001$). Although not specific to pharmacology, GPT-4.0 and GPT-4o had significantly ($p < 0.05$) greater correct answers for clinical management scenarios that medical students
Bland. 2025 United States of America	Enhancing medical student engagement through cinematic clinical narratives: Multimodal generative ai-based mixed methods study	To enhance clinical pharmacology teaching by using multimodal generative artificial intelligence to create cinematic, narrative-driven clinical learning materials	Cross-sectional study Medicine	Yes 18 students	Paedagogy	Chain-of-thought ChatGPT 4.0 Eleven Labs# Leonardo. a# Stable Diffusion# Suno#	Students were positive towards the cinematic clinical narrative in various dimensions (4.28 to 4.72), including satisfaction, engagement, usefulness, and learning. Students preferred the revised version (77.8%) in comparison to those that had no preference (16.7%) or preferred the traditional format (5.6%). Students remarked that the revised case was more engaging, entertaining, and memorable, however, suggested more interactive and detailed learning to be included, as well as technical refinement
Cai and Tung. 2025 New Zealand	ChatGPT-4's capability in addressing multiple-choice questions within the primary examination of the Australian and New Zealand College of Anesthetists	To appraise ChatGPT-4's accuracy on a substantially expanded set of official multiple-choice questions	Cross-sectional study Medicine	No 59 pharmacology questions from the Australian and New Zealand College of Anesthetists primary examination	Assessment	Zero-shot ChatGPT 4.0	ChatGPT 4.0 scored 76% for the pharmacology component of the test, which was in alignment to physiology scores (85%), but higher than clinical measurement (63%) and anatomy (0%). However, this comparison is unequal give the larger focus on physiology and pharmacology. ChatGPT 4.0 passed the standard setting for the score threshold. At times, ChatGPT 4.0 provided nuanced explanations for the correct or incorrect answers
Cherif et al. 2024 Tunisia	Appraisal of ChatGPT's aptitude for medical education: Comparative analysis with third-year medical students in a pulmonology examination	To evaluate and compare ChatGPT's performance on a pulmonology examination against third-year medical students	Cross-sectional study Medicine	Yes (244 third-year medical students) 45 questions from the Faculty of Medicine Health of Tunis question bank	Assessment	Zero-shot and chain-of-thought ChatGPT 3.5	Only 43.9% of students passed the examination, with a median score of 48.9%. ChatGPT-3.5 scored 46.7% when not contextualized, and 51.1% when contextualized (which allowed it to achieve a passing score). For pharmacology, students scored 70%, and the decontextualised and contextualized ChatGPT-3.5. versions scored 60% and 70%, respectively. Although not specific to pharmacology, ChatGPT-3.5 struggled more with independent multiple-choice and short open-ended questions, and specifically when clinical cases were incorporated. Similarly, not considering pharmacology specifically, ChatGPT-3.5 performed better in low- or intermediate-difficulty questions, but less so than students in higher-difficulty questions

(Continues)

TABLE 1 | (Continued)

Authors, year and country	Title	Aim of study ^a	Study design and discipline or profession	Inclusion of students in experimentation, and total number of participants or samples	Focus of study	Artificial intelligence tool	Findings of the study
Choi. 2023 South Korea	Assessment of the capacity of ChatGPT as a self-learning tool in medical pharmacology: A study using MCQs	To determine ChatGPT's effectiveness as a standalone self-learning tool by assessing its multiple-choice questions accuracy and the credibility of its rationales	Cross-sectional study Basic sciences and Medicine	No 312 multiple-choice questions derived from 78 test items from the Korean Comprehensive Basic Medical Sciences Examination	Assessment	Zero-shot and chain-of-thought ChatGPT#	ChatGPT scored 71.8% for the test items, with a 76% accuracy between the increasing prompt type (correct answer, rationale, reference, relevant paragraph). It was best to answer recall questions (81.8%), then interpretation (75.0%) and lastly problem-solving (35.7%). Of the items, there was a 62.8% correct concordance, 14.1% incorrect concordance, and 23.1% partially correct discordance between prompts. Correct answers were supported by scientifically accurate information (73.5%; $p < 0.05$), while incorrect answers were supported by scientifically inaccurate information (19.6%; $p < 0.05$). Although not significant, correct answers were supported by paragraphs (46.2%), while 25.6% were not. Incorrect answers were supported by paragraphs (14.1%), while 14.1% were not. Significant ($p < 0.05$) association between correct answers and relevant paragraphs was seen (75.4%), while incorrect answers were supported by correct but irrelevant paragraphs (9.8%) or incorrect paragraphs (8.2%). Three errors were noted for references, including reference URLs not found or erroneous (69.7%), authors not being present (16.9%), and incorrect titled references (97.4%)
Culp et al. 2024 United States of America	An artificial intelligence-supported medicinal chemistry project: An example for incorporating artificial intelligence within the pharmacy curriculum	To integrate AI tools into a medicinal chemistry course to teach core concepts and increase pharmacy students' familiarity with artificial intelligence in practice and drug development	Pre-post interventional study Pharmacy	Yes (83 to 161 participants [pre vs. post])	Paedagogy	Zero-shot KNIME	The student's familiarity with artificial intelligence was not significantly altered by the intervention (68.7% vs. 77.1%), however, their familiarity with the application of artificial intelligence in pharmacy was increased (25.3% to 64.3%), as well as their software use (24.1% to 64.3%) and interpretation of such results (16.9% to 53.5%) (all significant $p < 0.05$)
Danesh et al. 2024 United States of America	Artificial intelligence in dental education: ChatGPT's performance on the periodontic in-service examination	To evaluate ChatGPT's foundational knowledge in periodontology using American Academy of Periodontology in-service exam questions	Cross-sectional study Dentistry	No 311 multiple-choice questions from the 2023 Periodontics In-Service Written Examination (34 pharmacology and therapeutics questions)	Assessment	Zero-shot ChatGPT 3.5 ChatGPT 4.0	A total of 91.2% and 94.1% correct answers were observed for ChatGPT 3.5 and ChatGPT 4.0, respectively. This was higher than other disciplines, including embryology and anatomy, microbiology, immunology, periodontal etiology and pathogenesis, diagnosis, treatment planning and prognosis, therapy and oral pathology/medicine (33.3% to 91.9%). Although not specific to pharmacology, 54.0% and 76.2% of responses had associated explanations for ChatGPT 3.5 and ChatGPT 4.0, respectively, where 61.1% and 76.9% were correct answers with explanations, and 44.3% and 74.4% were incorrect answers with explanations. Performance may have been better for questions with lower level on insight

(Continues)

TABLE 1 | (Continued)

Authors, year and country	Title	Aim of study ^a	Study design and discipline or profession	Inclusion of students in experimentation, and total number of participants or samples	Focus of study	Artificial intelligence tool	Findings of the study
Driesnack et al. 2024 Germany	A practice-based approach to teaching antimicrobial therapy using artificial intelligence and gamified learning	To assess how gamified learning affects factual antibiotic-prescribing knowledge and uncertainty, and to explore artificial intelligence-based evaluation of freeform student responses	Pre-post interventional study Medicine	Yes (18 medical students) 10 Freeform questions	Assessment	Zero-shot GPT-4-0125-preview	Of a subset of 50 answers, human reviewers agreed (92%, Cohen's kappa = 0.7) that large-language models' feedback on answers were helpful to students' learning. No incorrect information was generated. Student answers were mostly listed as C (not entirely aligned to expert answer) by the large-language model, however, 8.89% and 15.56% were listed as A and B (higher alignment with expert answers). Some were ranked as D (74.44%), suggesting disagreement with expert answers. Human reviewers had fair to moderate agreement (Cohen's kappa 0.36 and 0.42) for labels A to E. The large-language model could generate individual reports on student learning and areas of remediation
Guillen-Grima et al. 2023 Spain	Evaluating the efficacy of ChatGPT in navigating the Spanish medical residency entrance examination (MIR): Promising horizons for ai in clinical medicine	To measure the performance of GPT-3.5 and GPT-4 in passing Spain's MIR medical specialist entrance examination	Cross-sectional study Medicine	No 182 questions from the Spanish Medical Residency Entrance Examination	Assessment	Chain-of-thought ChatGPT 3.5 ChatGPT 4.0	Both Spanish and English version of the test were answered correctly by ChatGPT-3.5 and 4.0 by 86.81%/87.91% and 63.18%/66.48%, respectively. Significant differences between ChatGPT versions were noted ($p < 0.001$), but not between languages. No difference were seen between theoretical (87.1%) and practical (86.6%) questions. ChatGPT 4.0 performed very poor in terms of pharmacology (40%), while most other disciplines were scored high in, apart from infectious diseases, forensics and legal medicine, pneumonology, and critical care. Consistency was present between iterative attempts to answer questions (Cohen's kappa 0.7418). Length of questions and amount of words didn't influence correctness. A 25% risk of error and harm was indicated, with 33.3% error, no harm using the NCC MERP classification system. Errors spanned the A to G categories, with most present as A (capacity to cause error), C (error reached patient by no harm), and D (error reached patient and required monitoring). None were at a stage of death risk. No harm were present when pharmacology questions were answered incorrectly. Pharmacology errors were thought to be linked to historical prevalence of certain guidelines, or incorrect balancing between adverse effects
Huang et al. 2024a Taiwan	Performance of ChatGPT on Stage 1 of the Taiwanese medical licensing exam	To evaluate ChatGPT's accuracy and logical reasoning across all 10 subjects of Taiwan's Senior Professional and Technical Examinations for Medical Doctors Stage 1 exam in both Chinese and English	Cross-sectional study Medicine	No 600 questions from the Taiwan medical licensing exam	Assessment	Zero-shot ChatGPT 4.0	Between three separate examination sittings (February, July, February), pharmacology had 80.7%, 96.0% and 96.0% correct answers, respectively, with 93.2% on average. On average, ChatGPT 4 scored very well between disciplines

(Continues)

TABLE 1 | (Continued)

Authors, year and country	Title	Aim of study ^a	Study design and discipline or profession	Inclusion of students in experimentation, and total number of participants or samples	Focus of study	Artificial intelligence tool	Findings of the study
Huang et al. 2024b Taiwan	Using machine learning to identify key subject categories predicting the pre-clerkship and clerkship performance: 8-year cohort study	To identify pre-clerkship subject categories that best predict later clerkship performance	Cross-sectional study Medicine	No 1162 students data	Programme evaluation	None Machine learning algorithms generated in Python 9.1	Pharmacology had an area under the curve predicting clerkship performance of 0.922. Machine learning techniques incorporating support vector machine had the best accuracy (0.871) in the basic sciences category. Pharmacology outcomes impacted clerkship performance, inclusive of other disciplines
Quah et al. 2024 Singapore	Performance of large language models in oral and maxillofacial surgery examinations	To determine the accuracy of large language models in answering oral and maxillofacial surgery multiple-choice questions	Cross-sectional study Dentistry	No 259 questions from the Discipline of Oral and Maxillofacial Surgery, Faculty of Dentistry, National University of Singapore question bank	Assessment	Chain-of-thought ChatGPT 3.5 ChatGPT 4.0 Copilot# Gemini# Llama 2#	The percentage of correct answers differed greatly between models ($p < 0.001$), with the best scoring model being Copilot (82.4%) and ChatGPT 4.0 (70.6%). All other models scored lower, with Gemini (29.4%), ChatGPT 3.5 (29.4%) and Llama 2 (17.6%) achieving poor scores. Similar discrepancies were noted for other disciplines. ChatGPT 4.0 and Copilot had the best overall scores with 76.8% and 72.6%
Roganović. 2024 Serbia	Familiarity with ChatGPT features modifies expectations and learning outcomes of dental students	To investigate how reading ChatGPT's feature descriptions in fluences willingness to use AI and affects dental students' learning performance	Cross-sectional study Dentistry	Yes 104 third-year students (13 used ChatGPT and read the description of ChatGPT, 47 read the description of ChatGPT but did not use ChatGPT, 44 did not use either)	Paedagogy	Not stated ChatGPT 3.5	Those who opted for ChatGPT had an initial higher expectation towards its use (44.0 vs. 34.5, $p < 0.05$), and expected that it would help them (4.0 vs. 3.3, $p < 0.05$). Artificial intelligence-associated risk assessment had no significant difference (3.3 vs. 2.7). After reading the feature description, those who did not use ChatGPT had an alteration of expectations: effort of ease (2.9 to 3.9), risk (2.7 to 3.6) and assistance (3.3 to 3.1). Similarly, those who used it changed their scores from 4.0 to 3.6 for expectations towards assistance. Pharmacology quiz scores was significantly higher ($p < 0.01$) in those that used and did not use ChatGPT, but used its figure description. Belief that ChatGPT was easy to use, that it was a new experience, and useful, was shown to affect quiz scores in those that did not use it. No association was noted for those that used it

(Continues)

TABLE 1 | (Continued)

Authors, year and country	Title	Aim of study ^a	Study design and discipline or profession	Inclusion of students in experimentation, and total number of participants or samples		Focus of study	Artificial intelligence tool	Findings of the study
Sridharan and Sequeira, 2024 Bahrain	Artificial intelligence and medical education: Application in classroom instruction and student assessment using a pharmacology & therapeutics case study	To evaluate the usefulness of three artificial intelligence platforms in identifying pharmacology-related learning objectives for hypertension management across medical-training levels	Cross-sectional study Medicine	No 5 questions	Assessment Curriculum design	Chain-of- thought ChatGPT 3.5 Claude- Instant# Sage Poe#	All three large-language models designed specific learning outcomes at pre-clerkship and at-graduation levels that were appropriate and aligned with what should be taught for hypertension management, and followed published guidelines. A-type multiple-choice questions were diverse, though each large-language model tended to have a different focal point for a few sections. Content validation was needed as some questions had contentious nuances. Commons errors included item construction defects, more than one correct answer, and inappropriateness for the student level. Claude-Instant had the lowest prevalence of defects, with none detected at-graduation level. The large-language models differed in guidance on standard setting, with Claude-Instant stating it could not provide simulated figures. For integrated case clusters, short-answer questions and objective structure practical examinations, deficiencies were noted for technical accuracy, comprehensiveness, education level and construction defects between all three models, however, Sage Poe did not generate any objective structure practical examinations. Claude-Instant was most free of errors	
Suchman et al. 2023 United States of America	Chat generative pretrained transformer fails the multiple-choice American College of Gastroenterology self-assessment test	To assess ChatGPT's accuracy on a gastroenterology self-assessment test	Cross-sectional study Medicine	No 300 questions from the American College of Gastroenterology self-assessment tests	Assessment	Zero-shot ChatGPT 3.0 ChatGPT 4.0	Stratification for pharmacology was not noted, but 65.1% and 62.4% of the questions were answered correctly by ChatGPT 3.0 and 4.0, respectively. In-text authors suggest that pharmaceutical regimens were all answered incorrectly	
Svensden et al. 2024 Norway	Short-term learning effect of ChatGPT on pharmacy students' learning	To investigate the immediate short-term learning effect of ChatGPT use among pharmacy students	Experimental randomized study Pharmacy	Yes Undergraduate and postgraduate pharmacy students (31 students)—15 intervention and 16 control	Paedagogy	Not stated ChatGPT 4.0	Post-test scores increased for the intervention (5.4 to 7.7) and control groups (4.6 to 5.9), however, there was a statistically significantly ($p=0.041$) higher score for the intervention group at post-intervention stage. However, when adjusting for pre-test scores, the difference in post- test scores were not significant anymore. Although response time decreased for the intervention group (8.5), it was not statistically significantly lower than the control group (9.9)	
Wang et al. 2023 Taiwan	Performance of ChatGPT on the pharmacist licensing examination in Taiwan	To examine ChatGPT's accuracy on Taiwan's Pharmacist Licensing Examination and evaluate its potential educational applications in pharmacy training	Cross-sectional study Pharmacy	No 313 questions from the Taiwanese Pharmacist Licensing Examination	Assessment	Zero-shot ChatGPT 3.5	For Chinese questions, ChatGPT scored 70.6%, 51.2%, 40.6%, 57.5%, 53.8%, and 50.0%, respectively for pharmacology and pharmaceutical chemistry, pharmaceutical analysis and pharmacognosy, pharmaceuticals and biopharmaceutics, dispensing pharmacy and clinical pharmacy, therapeutics, and pharmacy administration and pharmacy law. For English translations, scores were 72.1%, 54.9%, 43.8%, 75.0%, 73.8% and 54.0%	

^aStudy aims have been standardised for readability.

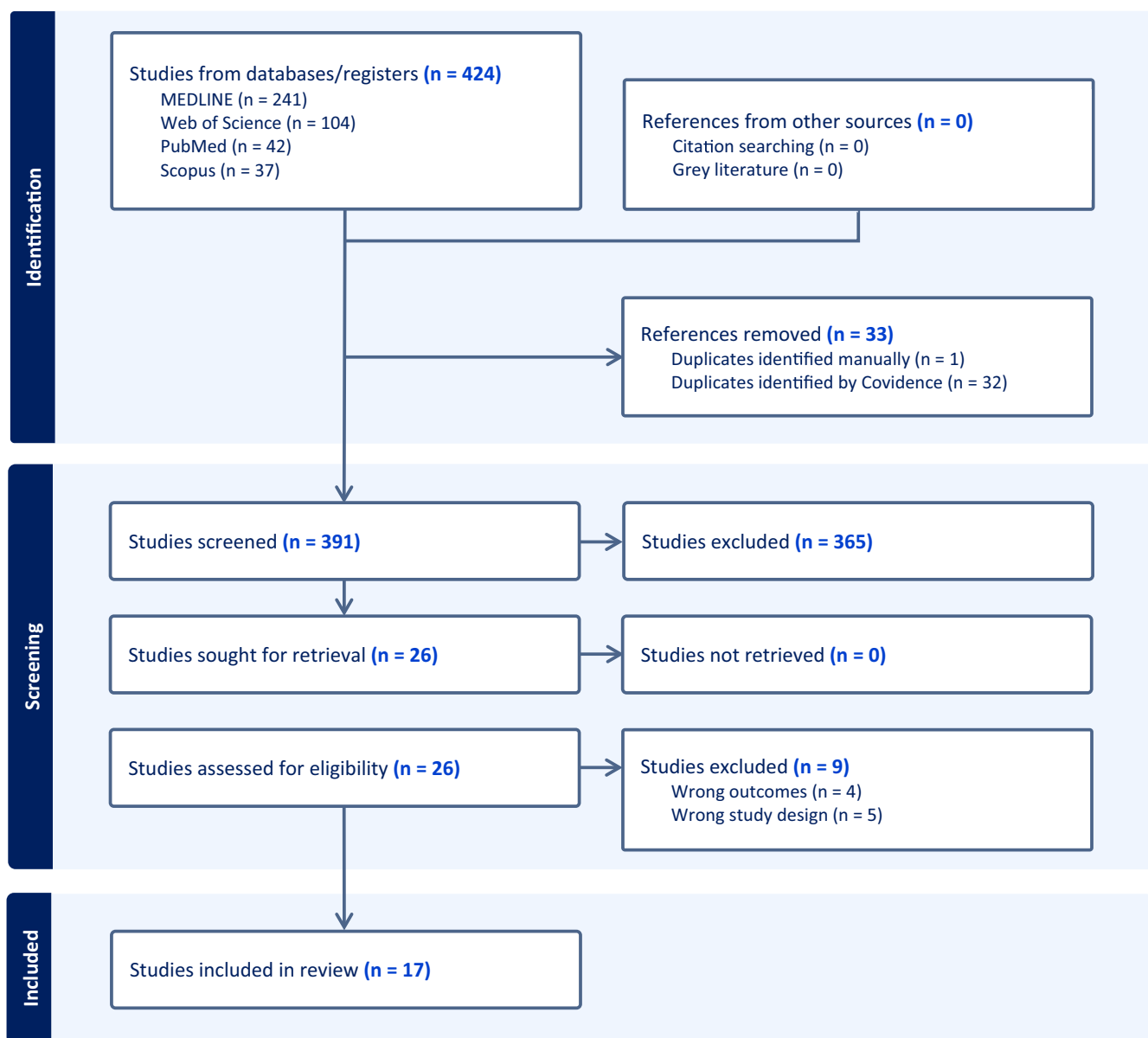


FIGURE 1 | PRISMA diagram.

South Korea, and Spain. Two studies were from middle-income countries: Serbia and Tunisia. No studies from low-income countries were identified. Most studies were within the past 2 years, which is unsurprising considering the recency of artificial intelligence developments. Studies were predominantly cross-sectional studies ($n = 14$), with only two pre-post interventional studies. Contextually, artificial intelligence use in pharmacology education has not received overt attention in literature, however, it is not surprising considering the relative newness of the field with recent advances, and likely to change in coming years. However, much of the research that is currently available is relatively preliminary, thus limiting potential for evidence-based practice for pharmacology itself.

The predominant setting for pharmacology education was in medicine ($n = 11$), with three studies centering around dentistry and pharmacy each, and only one incorporating pharmacology as a basic science. Considering the importance of pharmacology in medicine, it is understandable that it has a predominant

feature in literature. However, contextual application cannot be undermined in different professions where the curricular lens and approaches may differ given unique outcomes and scope of practice of different practitioners. For example, while physicians require prescribing competencies [22], pharmacists at a graduate level typically do not prescribe; instead they approach pharmacology from the perspective of dispensers and providers of pharmaceutical care [36]. As such, the learning approach may differ significantly depending on not only what needs to be developed, but also in which context it needs to be taught for the most authentic practice. By generating a diversity of contextually relevant evidence, the potential for purposeful inclusion into curricula becomes grounded in explicit scholarship, decreasing the risk for inappropriate application, especially when implemented by educators with less understanding of curriculum development and instructional design.

Artificial intelligence was mostly evaluated in terms of assessment use ($n = 12$), with four investigations as a pedagogical

TABLE 2 | Primary characteristics of the study design, participants, domains, and artificial intelligence tools used.

Parameter	Type	Number (%)
Study design	Cross-sectional study	14 (82.4)
	Experimental randomized study	1 (5.9)
	Pre-post interventional study	2 (11.8)
Disciplinary/professional focus of pharmacology ^c	Basic sciences	1 (5.9)
	Dentistry	3 (17.6)
	Medicine	11 (64.7)
	Pharmacy	3 (17.6)
Students or academic staff part of the experimentation	Yes	6 (35.3)
	No	11 (64.7)
Domain artificial intelligence used in ^c	Assessment	12 (70.6)
	Curriculum design	1 (5.9)
	Paedagogy	4 (23.5)
	Programme evaluation	1 (5.9)
Prompting strategy used ^{a,c}	Not stated	2 (11.8)
	Zero-shot	10 (58.8)
	Chain-of-thought	6 (35.3)
<i>Tools^c</i>		
Text-generation ($n=26$)	ChatGPT ^b	1 (5.9)
	ChatGPT 3.0	1 (5.9)
	ChatGPT 3.5	8 (47.1)
	ChatGPT 4.0	9 (52.9)
	ChatGPT 4o	1 (5.9)
	Claude-Instant ^b	1 (5.9)
	Copilot ^b	1 (5.9)
	Gemini ^b	1 (5.9)
	ChatGPT-4-0125-preview	1 (5.9)
	Llama 2 ^b	1 (5.9)
	Sage Poe ^b	1 (5.9)
	Eleven Labs ^b	1 (5.9)
Audio generation ($n=2$)	Suno ^b	1 (5.9)
	Leonardo.a ^b	1 (5.9)
Visual generation ($n=2$)	Stable Diffusion ^b	1 (5.9)
	KNIME ^b	1 (5.9)
Workflow/application ($n=2$)	Machine learning algorithms generated in Python 9.1	1 (5.9)

^aMachine learning was used in a single study, which does not make use of a prompting strategy.^bVersion not disclosed.^cSome studies investigated multiple disciplinary/professional settings, domains, prompting strategies, and tools.

tool, and a single occurrence for curriculum design and programme evaluation each. Artificial intelligence tools were used in general for text-generation ($n=16$), though audio ($n=2$) and visual generation ($n=2$) were present as well, including

their use as an application for modeling or workflow ($n=2$). The format of artificial intelligence used was overtly generative artificial intelligence ($n=15$), where machine learning and artificial intelligence-augmented platforms were present

as single occurrences each. ChatGPT was the most often-used tool ($n=21$), particularly ChatGPT 3.5 ($n=8$) and 4.0 ($n=9$). Others included (as one instance each): Claude Instant, 11 Labs, Gemini, KNIME, Leonardo.a, Llama 2, machine learning generated in Python, Sage Poe, Stable Diffusion and Suno. Prompting strategies overtly made use of zero-shot ($n=10$) and chain-of-thought prompting ($n=6$). Most studies did not involve students in their experimental design ($n=11$), focusing more on the artificial intelligence model itself than its use by students in their learning.

3.2 | Quality and General Confounding Considerations

In terms of quality (Table 3), all publications had clear inclusion criteria, reliable exposure measurements, and objective standard criteria for measurement; the majority adhered to clear study subjects and settings, valid outcome measurements, and appropriate statistical methods. However, confounding factors were identified and mitigated in only 35.39% of cases, which highlights concerns of reproducibility, reliability, and impact measurement that may be yielded from it. Generalized concerns are described below, while those that are more relevant within the context of the described literature have been discussed in Section 3.3 for clarity.

3.2.1 | Cross-Sectional Study Design

Given the overt cross-sectional study design, several limitations and confounders must be acknowledged. The longitudinal impact of artificial intelligence on pharmacology education is unavailable, creating a reduced understanding of the effects on competency development and attainment of graduate outcomes. For example, generative artificial intelligence-produced cinematic clinical narratives [5], ChatGPT-supported learning [3], or KNIME-based projects in medicinal chemistry [17] were assessed as single iterations, limiting an understanding of whether perceptions, interest or knowledge attainment would extend beyond an initial period, or fair better than standard modalities. Additionally, given the expansion of generative artificial intelligence models, the strength of newer reasoning models, such as ChatGPT 5.4, is likely to afford a much greater outcome than prior versions (e.g., ChatGPT 3.0). Two comparative studies highlighted an improved assessment response by newer ChatGPT versions [4, 10], showcasing that longitudinal studies of different versions, or even within the same model across a time period, will alter efficiency.

3.2.2 | Prompt Engineering

Prompt engineering provides a structured approach for human-defined instructions to generative artificial intelligence tools to facilitate a specific process, and thus is imperative for success [37, 38]. Zero-shot prompting, the predominant practice seen in the study, provides little information, context or outcome expectations, which precipitates less clear outcomes. More complex prompting strategies, such as few-shot [37] or chain-of-thought prompting [33] afford the model more examples of the expected

outcomes, or the contextual and thinking processes necessary into the prompt. The included studies did not highlight or discuss prior training of individuals on prompt engineering or use of artificial intelligence. It is understandable that background and expertise would not inherently be mentioned by authors, and thus an a priori assumption of appropriate use is considered. That being said, prompting instructions were discussed at various levels, ranging from superficial to more explanatory. Prompting quality differs greatly between individuals; thus, care should be taken in how it may bias the outcome of research, which invariably may alter the comparability or applicability in practice. As such, poor or inefficient prompting, including individual variability among educators, may yield incomparable strategies, dissuade individuals from further use, or exacerbate complications in its application in education.

The basic zero-shot or non-directed prompting applied in the majority of studies creates an inherent interpretative limitation, where the potential accuracy of the generative artificial intelligence model may be underestimated based on unclear instructions. For example, contextualized or chain-of-thought prompts proved more successful in answering assessments than basic prompts [7], highlighting that poorer outcomes may inherently be obtained if the underlying experimental method did not consider it within their design. Similarly, given the inherent variability of prompting between individuals, use of generative artificial intelligence for learning by students without clear structure in experimental paedagogic research [3, 18] creates a heterogeneous environment where learning becomes subject to the students' ability to prompt effectively, rather than whether the generative artificial intelligence itself can promote it. Particularly when considering student involvement, the limited reporting on whether baseline training was provided prior to use introduces a high risk of bias and variability based on user background. Although not directly related to prompt engineering itself, Roganović [3] did show that student exposure to information on ChatGPT features (broad use, strengths and limitations) altered their use. In support, a relationship has been observed between the quality of prompt engineering and the outcome provided to students [39–41].

3.2.3 | Instructional Design, Rationale and User Involvement

As mentioned in Section 3.1, literature was on assessment ($n=11$) and pedagogy ($n=4$), though considering heterogeneity, care must be taken when interpreting confounders or educational reach. Regarding generative artificial intelligence models' ability to answer question banks and licensure examinations, a general observation was the limited downstream rationale for educational incorporation. Clarity of how this would be incorporated, whether as personalized formative learning systems, self-regulated questioning platforms, quality control mechanisms, potential of hindering non-invigilated examinations, or as exploratory research on the models' capabilities, was low. As such, one needs to consider the value or reach of such studies should contextualisation not inherently be clear to suggest feasible guidance or implementation. To support instructional design, the underpinning scholarship is thus necessary to ensure rationale use.

TABLE 3 | Quality assessment of eligible articles.

Authors, year and country	Title	Clear inclusion criteria	Clear study subjects and setting	Valid and reliable exposure measurement	Objective, standard criteria for measurement	Confounding factors identified	Confounding factor mitigation identified	Valid and reliable outcome measurement	Appropriate statistical methods
Bicknell et al. 2024 United States of America	ChatGPT-4 omni performance in USMLE disciplines and clinical skills: Comparative analysis	Yes	Yes	Yes	Yes	Unsure	Unsure	Yes	Yes
Bland. 2025 United States of America	Enhancing medical student engagement through cinematic clinical narratives: Multimodal generative ai-based mixed methods study	Yes	Yes	Yes	Yes	No	No	Yes	Yes
Cai and Tung. 2025 New Zealand	ChatGPT-4's capability in addressing multiple-choice questions within the primary examination of the Australian and New Zealand College of Anesthetists	Yes	Yes	Yes	Yes	Unsure	Unsure	Unsure	No
Cherif et al. 2024 Tunisia	Appraisal of ChatGPT's aptitude for medical education: Comparative analysis with third-year medical students in a pulmonology examination	Yes	No	Yes	Yes	No	No	Yes	Yes
Choi. 2023 South Korea	Assessment of the capacity of ChatGPT as a self-learning tool in medical pharmacology: A study using MCQs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Culp et al. 2024 United States of America	An artificial intelligence-supported medicinal chemistry project: An example for incorporating artificial intelligence within the pharmacy curriculum	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No

(Continues)

TABLE 3 | (Continued)

Authors, year and country	Title	Clear inclusion criteria	Clear study subjects and setting	Valid and reliable exposure measurement	Objective, standard criteria for measurement	Confounding factors identified	Confounding factor mitigation identified	Valid and reliable outcome measurement	Appropriate statistical methods
Danesh et al. 2024 United States of America	Artificial intelligence in dental education: ChatGPT's performance on the periodontic in-service examination	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Driesnack et al. 2024 Germany	A practice-based approach to teaching antimicrobial therapy using artificial intelligence and gamified learning	Yes	Yes	Yes	Yes	Unsure	Unsure	Yes	Yes
Guillen-Grima et al. 2023 Spain	Evaluating the efficacy of ChatGPT in navigating the Spanish medical residency entrance examination (MIR): Promising horizons for ai in clinical medicine	Yes	Yes	Yes	Yes	Unsure	Unsure	Yes	Yes
Huang et al. 2024a Taiwan	Performance of ChatGPT on Stage 1 of the Taiwanese medical licensing exam	Yes	Yes	Yes	Yes	No	No	Yes	Yes
Huang et al. 2024b Taiwan	Using machine learning to identify key subject categories predicting the pre-clerkship and clerkship performance: 8-year cohort study	Yes	Yes	Yes	Yes	No	No	Yes	Yes
Quah et al. 2024 Singapore	Performance of large language models in oral and maxillofacial surgery examinations	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Not applicable
Roganović. 2024 Serbia	Familiarity with ChatGPT features modifies expectations and learning outcomes of dental students	Yes	Yes	Yes	Yes	No	No	Yes	Yes

(Continues)

TABLE 3 | (Continued)

Authors, year and country	Title	Clear inclusion criteria	Clear study subjects and setting	Valid and reliable exposure measurement	Objective, standard criteria for measurement	Confounding factors identified	Confounding factor mitigation identified	Valid and reliable outcome measurement	Appropriate statistical methods
Sridharan and Sequeira. 2024 Bahrain	Artificial intelligence and medical education: Application in classroom instruction and student assessment using a pharmacology & therapeutics case study	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Suchman et al. 2023 United States of America	Chat generative pretrained transformer fails the multiple-choice American College of Gastroenterology self-assessment test	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Svendsen et al. 2024 Norway	Short-term learning effect of ChatGPT on pharmacy students' learning	Yes	Yes	Yes	Yes	No	No	Yes	Yes
Wang et al. 2023 Taiwan	Performance of ChatGPT on the pharmacist licensing examination in Taiwan	Yes	Yes	Yes	Yes	No	No	Yes	Yes

Included studies lacked clearly indicated underpinning frameworks or scholarships, though exceptions were evident. Bloom's taxonomy, for example, was used by Sridharan and Sequeira [12] to help produce learning outcomes and assessments, thus allowing for a more targeted alignment with learning expectations. Similarly, Bland [5] incorporated constructivist and Mayer's cognitive theory of multimedia learning to support the design of cinematic clinical narratives, highlighting a clearer design platform. The Technological Paedagogical Content Knowledge (TPACK) [25] and Passive, Interactive, Creative, Replacement, Amplification, Transformation (PICRAT) [42] frameworks, which provide guidance on how to effectively, rationally and purposefully apply digital tools, are often proposed to mitigate irrational application of technology. Bland [5], in support of clear design, implemented TPACK to align technology, pedagogy and content, promoting a cohesive approach to integrate all three knowledge spheres. Beyond such examples, application of artificial intelligence was not reported as guided by a specific framework, and thus, assessment of their underlying alignment to such frameworks is not possible without assumptive bias. Given the limitation, one can position most studies though at a passive replacement level (according to PICRAT); while studies such by Bland [5], Culp et al. [17], Roganović [3] and Svendsen et al. [18] approached more interactive amplification.

The limited use of students highlights that artificial intelligence was the focal point of the research, and underscores a prominent limitation and area of future research. Although the publishing of research on the artificial intelligence models itself is important, the incorporation of the student as downstream user allows for potential impact determinations, for example, their perceptions, attitudes, knowledge improvement, or behavioural changes. Without such determinations, the potential effect of artificial intelligence remains theoretical, decreasing the evidence supply for practical implementation and best-practice guidance. Similarly, educators who drive its incorporation into curricula would be important to include as stakeholders, which was also not seen in many of the included studies. Where involved, such as with Choi [8], Driesnack et al. [9], and Sridharan and Sequeira [12], the importance of human-supported vetting was highlighted to promote error-vetting, feedback quality, and answer alignment.

3.3 | Themes of Artificial Intelligence Use in Pharmacology Education

3.3.1 | Curriculum Design and Programme Evaluation

A single publication each focused on the use of artificial intelligence in curriculum design [12] and programme evaluation [14]. Sridharan and Sequeira [12] assessed three tools (ChatGPT 3.5, Claude Instant and Sage Poe) to develop a hypertension-focused session's pharmacotherapeutic learning outcomes at pre-clerkship and at-graduation level for a medical programme. Although comparable outcomes were generated which aligned to Bloom's taxonomy [12], tools displayed differences in outcome complexity, integration of clinical guidelines, explicitness of key concepts, and breadth of analysis of patient factors. Although relevant and feasible for use [12], care should be taken

in the constructive alignment to the broader programme level outcomes, its horizontal and vertical integration, and national needs. It is unclear whether contextual alignment to such broader elements was considered (e.g., the rest of the curriculum or scope of practice), though may invariably differ between the three tools used, and thus affect their compatibility with the programme itself.

Huang et al. [14] used machine learning models via Python to evaluate non-academic and academic factors predicting medical clerkship success, of which pharmacology was one such discipline (area under the curve of 0.922). Given the importance of pharmacology to the profession, it is not unexpected that such an association exists [14], and thus underscores the importance of clearly articulating its relevance to the field, placing emphasis in teaching and learning, and using such data to drive individualized learning and remedial pathways. Importantly, large datasets of fit-for-purpose data are needed to allow for such predictive models to be developed, which may not always be readily available or consistently applied [43]. Considering the development of the field of data science, artificial intelligence, such as that applied by Huang et al. [14] is an exciting field of interest which may provide insights into factors associated with success, or flagging of risk factors that allow for more personalized learning or remediation strategies to be applied.

3.3.2 | Assessment

For assessment, studies investigated the ability of generative artificial intelligence to answer pharmacology-related questions [1, 2, 4, 6–8, 10, 11, 13, 19], to generate pharmacology-related questions [12], and to provide feedback on student-answered questions [9]. Most studies report generative artificial intelligence systems' ability to pass such assessments [1, 2, 4, 6–8, 11], though failures have also been reported for pharmacy [19], general medicine [10] and gastroenterology [13]. Several studies suggest that generative artificial intelligence performed better in pharmacology than other disciplines in the profession [1, 2, 6, 19], however, this was not always the case [10]. Within dentistry, two studies showed that generative artificial intelligence could answer periodontic licensure [1] and oral and maxillofacial faculty question banks, where ChatGPT and Gemini appeared to outperform Copilot, Gemini and Llama 2 in the latter study [2]. Wang et al. [19] reported ChatGPT-3.5's mixed ability to answer Chinese and English versions of the Taiwanese Pharmacist Licensing Examination questions, ultimately failing the examination, although better scores were seen for pharmacology and pharmaceutical chemistry sections, and generally towards the English translation thereof. Seven studies reported mixed results in relation to medicine and sub-specialties thereof [4, 6–8, 10, 11, 13], with two studies including historical or current student comparator groups [4, 7]. Various versions of ChatGPT were assessed between studies (3.5, 4.0, and 4o), with results ranging from 40% to 96% correct answers [4–6, 10, 11, 13], with either better [4] or comparable [7] results to students. Several factors were associated with better performance, including later versions of ChatGPT [4, 10], contextualisation of prompts alongside questions [7], or easier questions [4, 7, 8]. One study highlighted ChatGPT-3.5's performance was better towards recall (81.8%)

and interpretation questions (75%) than problem-solving questions (35.7%) [8]. No association was seen though with language (Spanish versus English), length of questions, or amount of words [10]. In terms of information support, Choi [8] reported that ChatGPT supported correct and incorrect answers by scientifically accurate and inaccurate information, with additional errors noted for references, such as erroneous reference links, authors, or titles. Guillen-Grima et al. [10] used the National Coordinating Council for Medication Error Reporting and Prevention (NCC MERP) classification system and saw no risk of harm when pharmacology questions were answered incorrectly, and was likely due to historical prevalence of certain guidelines or incorrect balancing between adverse effects. As discussed in Sections 3.2.2 and 3.2.3, such studies provide valuable information on the models' capabilities, though without clear discussion of their educational rationale, the application within pharmacology education as a tool for teaching, learning and assessment is minimized.

Sridharan and Sequeira [12] evaluated ChatGPT-3.5, Claude Instant, and Sage Poe's ability to generate A-type and case cluster multiple-choice questions, integrated short-answer questions, and objective standardized practical examinations for antihypertensive pharmacology, as well as its ability to test standard-setting parameters. All three models generated questions with varying levels of success, showing convergence and divergence of focus points, such as drug type, pharmacokinetic aspect, and clinical scenario [12]. All three models also succumbed to a variety of quality deficits, including item construction defects, multiple correct answers, and appropriateness for the student level. Questions generated at graduation level tended to have less errors than those for pre-clerkship, with the number of errors differing between topics and tool [12]. Standard-setting was acknowledged by the tools as being purely hypothetical, either acknowledging their inability to do so or providing simulated findings [12]. The study highlights the potential for the use of such tools to generate assessments; however, it stresses the importance of expert-led quality control of the content and structure of the questions [12].

Driesnack et al. [9] used ChatGPT-4-0125-preview to generate feedback on whether students' answers on an antimicrobial freeform test aligned to expert answers, and which areas of knowledge deficit could be improved upon for future studies. Human reviewers had substantial agreement (Cohen's κ 0.70) that feedback was helpful, suggesting that with a sufficiently large sample for calibration, the need for human intervention could be reduced [9]. However, feedback was not circulated to students for their perceived benefit of remedial support, thus making it unclear how much contribution the end-user would find. This underscores the necessity of student involvement for rational and impactful educational use.

3.3.3 | Paedagogy

Four articles [3, 5, 17, 18] evaluated the paedagogic potential of artificial intelligence (of which three comprised generative artificial intelligence). Svendsen et al. [18] and Roganović [3] both assessed the alteration of knowledge outcomes after a generative artificial intelligence-based intervention, with

mixed results. Svendsen et al. [18] assessed the knowledge of Bachelor and Master of Pharmacy students before and after a 45-min study period, which was either with or without ChatGPT-4.0 support. Although no statistically significant improvement was noted after adjusting for pre-test scores, there was a trend for greater improvement in scores (likely undermined by the small sample size) [18]. Conversely, Roganović [3] measured the potential of learning more about ChatGPT-3.5 as a factor for altering outcomes when using or not using it for learning. Dentistry students were self-stratified into three groups: ChatGPT-use for learning after reading a description about the system; no use of ChatGPT after reading the description; or did not use ChatGPT nor read the description. Students scored better in an adverse effects quiz when using ChatGPT-3.5 than users who did not read the description or used ChatGPT ($p < 0.01$); however, no difference was seen in comparison to those that did not use ChatGPT but read the description [3]. Results suggest that students' expectations and awareness of how the system works and how to use it may bias their way of approaching their learning, thus indicating a potential relevance to incorporating it into their learning path [3]. For both studies, it should also be noted that students' use of ChatGPT for learning appeared to have been relatively or completely unstructured, and thus consideration is needed whether there was a purposeful application within the learning session or a student-determined approach to using it for learning. Learning design, including digital tools, should be tailored for activities, and thus the true impact may be confounded if not structured according to what is needed. Implementation of frameworks for instructional design, such as TPACK [25] and PICRAT [42], supports educational rationale and avoids the addition for luxury or novelty.

Both Bland [5] and Culp et al. [17] highlighted purposeful and design-supported implementation of artificial intelligence in learning design. Bland [5] determined the potential of a generated cinematic clinical narrative as an interactive multimedia learning platform. Several generative artificial intelligence models were used, including ChatGPT-4.0 (text), Leonardo.ai (images), 11 Labs (narration), and Suno (theme song, supported by ChatGPT-4.0), which allowed for the production of an immersive story, underpinned by the TPACK education framework [5]. Student feedback highlighted their positive experience (4.28 to 4.72 out of 5.00 for several indicators of satisfaction), with 14 out of 18 indicating a preference for such a modality [5]. Culp et al. [17] used KNIME, an artificial intelligence-augmented platform, in a first-year pharmacy medicinal chemistry course to help facilitate drug-protein interaction learning, with purposeful design of the learning opportunity to incorporate preparation and project activities without disrupting the key learning outcomes. Students indicated a growing understanding and application of artificial intelligence after engaging with the project and had a greater interest in the course content [17]. However, the impact of these learning opportunities towards objective changes within learning, whether as knowledge or more complex graduate outcomes, was not assessed, highlighting a stop-gap in the impact of reporting of such endeavors.

Furthermore, certain paedagogic studies [5, 18] were limited by a small sample size (15 to 18 students), creating potential room

for bias, weak statistical power, and thus undermining their reach. Three of the four pedagogic studies lacked an experimental, controlled format, hindering the true effect measurement of artificial intelligence's support [3, 5, 17].

3.4 | Ethical Considerations for Use in Pharmacology Education

Artificial intelligence introduces several important ethical considerations for use in education, including the potential impact on learning, its integration into education, and the appropriate use by stakeholders (such as students and educators) [44]. Artificial intelligence, as a tool, can deliver high-quality outputs; though, as explained previously, its quality will depend on how users apply it. Considering the importance of educational scholarships underpinning appropriate decision-making, ill-structured use may lead to irrational instructional design or student over-reliance for learning [45]. Discussion of accountability [34, 44] is thus needed, which invariably should rest with the user. This further positions the importance of transparency of use [34, 44], as it clarifies the curriculum and instructional design ecosystem. For example, Roganović showcased the change in use by students based on their understanding of generative artificial intelligence models [3], while Choi [8] and Sridharan and Sequiera [12] highlighted the prevalence of educational and content-based errors that exemplify the need for human involvement. Furthermore, contextual limitations are also apparent, where artificial intelligence may lack understanding of geographical and cultural diversity or localisation, thus creating outputs that may not be appropriate in all circumstances [34, 45].

Various assessment-related concerns may also arise. For example, when generating questions, clear structural guidelines and prompt engineering is needed [38], such as using a relevant knowledge base (e.g., programme-associated learning material or scope; best practice for assessment generation) or examples of outputs needed (e.g., structure of questions; flaws that should be checked for). Similarly, care should be taken to not place excessive trust in artificial intelligence, thus avoiding or undermining the human expert-factor necessary in vetting such questions [32, 45]. Ateeq et al. [28] recommends that in light of academic integrity concerns, the transition away from rote memorisation is needed, thus focusing on the development of authentic and comprehensive activities that are less likely to be impacted by artificial intelligence. One could argue that although subjective assessments in non-invigilated states, such as assignments [37], may be more at risk of artificial intelligence interference than those in invigilated settings, the mapping performed in this study showed that generative artificial intelligence could generally achieve passing grades in licensure examinations. As such, the potential impact of using generative artificial intelligence as a learning tool for such assessments may thus, in itself, raise similar concerns of whether it facilitates rote learning rather than higher-level competency development.

Considering the impact of pharmacology on patient safety, artificial intelligence use should also not undermine sound clinical decision-making, treatment integrity, or professional responsibility. An evidence-based approach is needed for safeguarding. Concerns have been raised about potential bias or algorithmic

determination of medication use and improper clinical decision-making [46, 47], thus necessitating judicious use and involvement of humans [47].

4 | Concluding Recommendations

The current mapping highlights the potential use of artificial intelligence in pharmacology education; however, it also indicates the infancy and preliminary nature of most research conducted. At the time of the study, most research, although adhering to quality indicators, did not stretch into higher-level outcomes such as the implementation of artificial intelligence in the curriculum, longitudinal changes in competency development, or feasibility thereof. This is underscored by the low number that included end-users, such as students, or suggested ways in which clear educational design principles could underpin its incorporation. This does not suggest the studies do not bear relevance to the field, but rather highlights areas of future research and ways of strengthening its impact. In general, a prominent consideration should be the strengthening of artificial intelligence-bolstered pharmacology education research to ensure impactful studies that promote clear evidence-based practice, thus affording readers more contextually relevant research and mitigating the dilution of the field's relevance and strength.

Based on the findings, several recommendations are proposed for the application of artificial intelligence in pharmacology education and associated research performed, though they can be extrapolated to other disciplines as well:

- a. Studies were, in general, limited and confounded due to lack of controls, study design limitations, or unclear reporting of artificial intelligence parameters. Studies which had a clear explanation of how prompting was performed and the models were selected, while including controls and comparative information, tended to provide a stronger evidence base. Reporting of artificial intelligence systems should be strengthened to include all necessary parameters, such as the version of the tool used, prompting strategy and associated prompts, excerpts of the feedback, time range, rationale for their selection, and where relevant, more complex settings such as the temperature. For example, checklists have already been developed that guide the minimum information needed when reporting artificial intelligence use in learning [48]. Given the exponential growth of the field, results may change quite frequently, thus impacting their reproducibility, the relevance to other readers, and the underlying assumptions or limitations of models, which reaffirms the importance of a clear understanding of what was present when used.
- b. Most studies lacked information on baseline expertise in artificial intelligence, particularly whether students were provided with sufficient training (although only four pedagogic studies were present). Generative artificial intelligence does not require broad expertise; however, good practice should still be followed which requires baseline understanding of use, including contextually relevant factors, strengths and weaknesses (potentially alongside more technically trained individuals), prompt engineering, and

ethical considerations. Generative artificial intelligence literacy [45] should be bolstered to ensure that users, whether students or educators, are sufficiently prepared to use models for the creation of material or facilitation of teaching, learning and assessment. For designers, the ability to access the backend of the models allows for further control over its outcomes, including temperature variation, promoting use. Higher education institutions are recommended to include basic training for students and educators on judicious and ethical artificial intelligence use to promote rational incorporation into education, whether via open-education resources or self-developed courses.

- c. Most studies resorted to zero-shot prompting, which itself confounds the potential outcomes as a human-subject consideration rather than the artificial intelligence model itself. As prompting strategies greatly affect the outcome, clear approaches are needed that define the contextual aspects needed and expectations, for example, using few-shot or chain-of-thought prompting. Poor prompting will inadvertently bias results and reduce their comparability or relevance.
- d. The majority of studies lacked involvement of students, arguably the most important end-user for education. By including academic staff and students, further understanding of impact is created. Beyond the invariable change management, user experience and perceptual aspects of artificial intelligence in education are important; otherwise, the research may remain theoretical with lower practical application if not clearly supported by data from human users. In doing so, educators or institutions may more clearly develop best-practice guidelines on the integration of artificial intelligence into curricula, and thus mitigate irrational or inappropriate use.
- e. Only two studies included a clear design strategy underpinned by frameworks. Integration of artificial intelligence in education requires careful evaluation and support of underlying educational principles and scholarship, including cognitive taxonomies [49] and instructional design frameworks, such as TPACK or PICRAT. In doing so, it allows for a structured way to justify and implement such technology. Furthermore, in particular for assessment design, guidelines are needed for the creation of high-quality and aligned questions, for example, using the NBME item writing guide [50]. Ultimately, artificial intelligence remains a tool that must be underpinned by educational scholarship and critical human involvement.
- f. Given the potential for epistemic bias from the sources of information for the tool, care should be taken to validate any outcomes generated for the specific context it will be applied in (e.g., country-specific essential medicine lists; programme level integration). As such, consideration should be given to clearly indicate such context in prompts, or provide sufficient knowledge for the tool, including relevant guidelines, associated curriculum aspects, and socio-cultural features. Content, and where possible, educational experts should ensure validation of all material is done

Artificial intelligence displays prominent potential for pharmacology education in a diversity of fields; however, care should be

taken to ensure that its application is supported by educational practice without disadvantaging its end-users. Continuous research, monitoring, and evaluation are needed to ensure the quality of its use in education is maintained, which will further contribute to evidence needed for its successful implementation within learning.

5 | Study Strengths and Limitations

The strength of the study includes its focus on pharmacology as a discipline, rather than broad disciplines that lose contextual relevance to the field. Although use of artificial intelligence may overlap broadly in various disciplines, its application is constrained by unique considerations of the discipline itself, such as its reliance on a myriad of prerequisite fundamental sciences, its translation into clinical applicability, and the high intrinsic load in comparison to other fields. Additionally, the study provides a unique perspective by evaluating the efficiency of generative artificial intelligence use in relation to its instructional design and prompt engineering strategies used, thus not merely highlighting current application, but also the rationality, purposefulness, technical proficiency, and educational strength of the study design. By providing focus on the prompting strategies used, it highlights potential bias, practice-related concerns, or inefficiencies that may affect the outcomes and educational potential. Furthermore, by assessing the quality in parallel to the literature itself, it highlights the preliminary nature of the work, while not undermining the impact thereof at its representative level. Additionally, it guides recommendations based on the gaps currently seen in the field, while integrating known educational frameworks. Although limited to relatively older models, the study provides a critical review of the purposeful implementation of artificial intelligence, including generative artificial intelligence, in pharmacology education, guiding further research design to allow for more scientific rigor and greater educational alignment.

Several limitations are noted in the analysis of the scoping review, including the limited number of publications, and particularly when stratifying according to the domain of study (such as paedagogy, curriculum design and programme evaluation). Given the exponential growth of generative artificial intelligence models, some research may already be less relevant than one would expect for other research fields; for example, ChatGPT-3.5 and ChatGPT-4.0 have already been replaced by newer versions with greater reasoning performance. No studies incorporating newer models, such as ChatGPT-5.0, are included given the period of investigation, which are likely to yield better results. Furthermore, the inter-rater reliability was fair, suggesting a potential reproducibility difficulty; however, this was mitigated through disagreement discussion relative to the eligibility criteria.

Author Contributions

Princess Ijeoma: investigation, writing – review and editing, methodology, formal analysis, validation, data curation. **Werner Cordier:** conceptualization, investigation, funding acquisition, writing

– original draft, methodology, validation, visualization, software, formal analysis, project administration, data curation, supervision, resources.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data will be made available on justified request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **File S1:** Search string.